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CO3

① Intelligent Agents in a Smart Home Automation System

1. Introduction

A smart home automation system uses Artificial Intelligence (AI) to control lighting, temperature, and security based on sensor inputs and user preferences.

2. Types of Intelligent Agent

1. Simple Reflex Agent

→ Acts based on current inputs and predefined rules.

→ Example: Turns ON lights when motion is detected.

→ Limitation: Cannot remember past events.

2. Model-Based Agent

→ Maintains information about the environment

→ Example: Tracks room occupancy and temperature

→ Advantage: Uses past and current information.

3. Goal-Based Agent

→ Selects actions to achieve specific goals.

→ Example: Maintains the desired room temperature

→ Advantage: Supports planning and decision-making.

4. Utility-Based Agent

→ Choose actions that provide the greatest overall benefit

→ Example: Balance comfort and electricity consumption

3. Comparison

Agent	function
simple reflex	Responds to current inputs
Model-based	Maintains Environmental information
Goal-Based	Achieves specific goals
utility-based	Balance multiple objectives

4. Best Approach

A hybrid approach is most effective. Model-based agents track the environment, goal-based agents maintain comfort and security, and utility-based agents reduce energy consumption.

5. Conclusion

A hybrid intelligent agent improves comfort, energy Efficiency, and security by combining fast responses, planning, and intelligent decision-making.

② Solving an unknown Maze using UCS and IDS

1. Introduction

A robot is an unknown maze can use uninformed Search algorithms to explore paths and find the exist

without prior knowledge of the layout.

2. Uniform cost search (UCS)

Steps:

1. start from the initial position
2. Explore the lowest-cost position with the lowest total path cost.
3. Add neighboring positions
4. Repeat until the exist is found.

Advantages :-

find the minimum-cost path when movements cost are non-negative

Disadvantages :-

Requires more memory and may explore many positions.

3. Herative Deepening Search (IDS)

IDS repeatedly searches with increasing depth limits until the Exist is found.

Advantages :-

uses less memory and finds the shallowest solution when all moves have equal cost.

Disadvantages :-

Repeats searches and does not gurantee minimum cost when movement cost differ.

Complexity:

- Time: $O(b^d)$
- Space: $O(bd)$

Here, b is the branching factor and d is the solution depth.

4. Comparison

Features	UCS	IDS
Based on	path cost	Search depth
memory	High	low
Best use	Different movement costs	Equal movements costs

5. Intelligent Agent in Maze Solving

- Simple reflex agent: Responds to nearby walls.
- Model-based agent: Remembers visited positions and walls.
- Goal-based agent: plans action to reach the exit
- utility-based agent: Considers distance, energy, and safety.

6. Best Approach

A model-based and goal-based Agent is suitable for an unknown maze. IDS is preferable when all moves have equal costs and memory is limited, whereas UCS is preferable when movement cost differ.

7. Conclusion

UCS finds minimum-cost paths, while IDS, saves memory for equal-cost paths. combining a model-based agent with a goal-based agent helps the robot explore the maze and reach the exist Efficiently.